

Python 3.13.7 (tags/v3.13.7:bcee1c3, Aug 14 2025, 14:15:11) [MSC v.1944 64 bit (AMD64)] on win32

Enter "help" below or click "Help" above for more information.

= RESTART: C:/Users/krish/AppData/Local/Programs/Python/Python313/Assessment 2.py

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QUESTION 1 - DOCTOR SHIFT SCHEDULING

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Valid Shift Schedule

D1 -> Morning

D2 -> Afternoon

D3 -> Night

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QUESTION 2 - ROBOT GRID NAVIGATION

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Shortest Path

(0, 0)

(1, 0)

(2, 0)

(2, 1)

(2, 2)

(3, 2)

(3, 3)

(3, 4)

(4, 4)

Total Cost = 8

Manhattan Distance from Start to Goal = 8

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QUESTION 3 - AUTONOMOUS RESCUE ROBOT

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Optimal Rescue Path

Base -> A -> D -> Victim

Minimum Cost = 7

Robot Actions

Scanning Area

Avoiding Obstacles
Finding Victim
Rescuing Victim
Sending Location to Rescue Team

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ALL PROGRAMS EXECUTED SUCCESSFULLY
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